

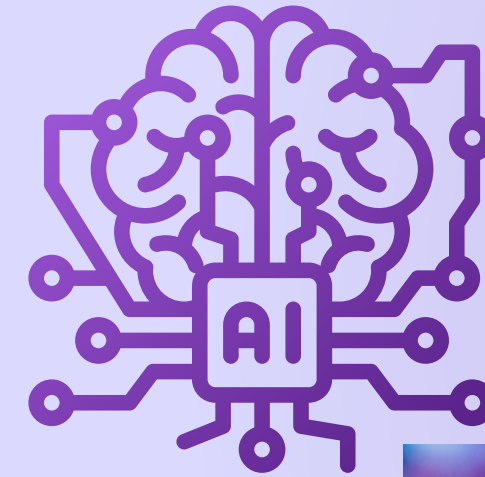
AI & DS PROJECT

BBC News Classification Model

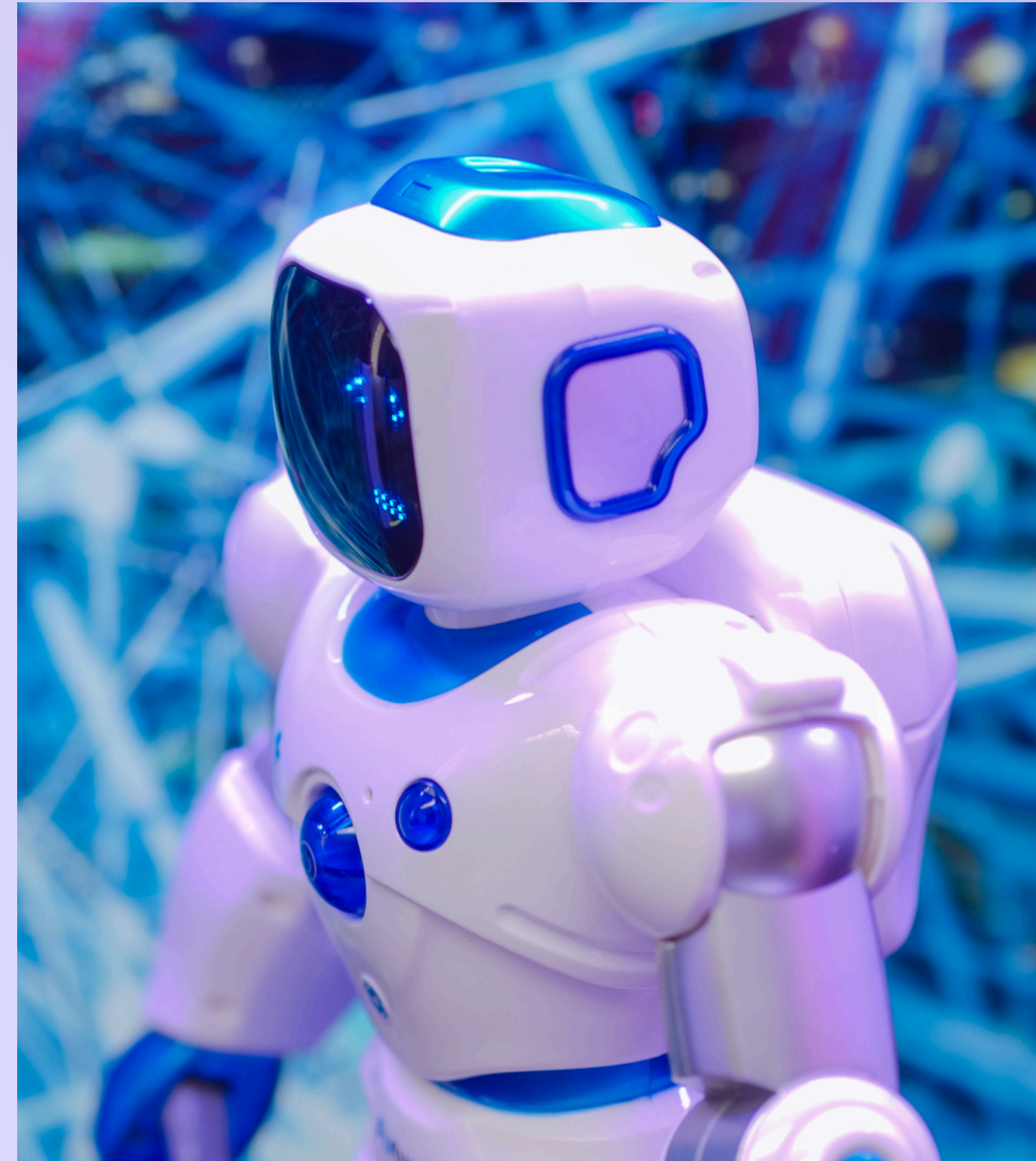
Subgroup - 04



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About Us

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Introduction

- Text classification is the process of assigning categories to textual data.
- The BBC News dataset contains well-labeled articles across five categories.
- We use Word2Vec to extract semantic meaning and classify news accurately.





Objective



- **Primary Goal:**

Classify BBC news articles into one of five categories: business, entertainment, politics, sport, or tech.

- **Approach:**

Use Word2Vec embeddings to convert textual data into dense semantic vectors, followed by classification using traditional machine learning algorithms (e.g., Logistic Regression, Random Forest).

- **Why Word2Vec?:**

Word2Vec captures semantic relationships between words by mapping them into continuous vector space. This allows the model to better understand context, synonyms, and topic relevance compared to basic Bag-of-Words or TF-IDF methods.



Dataset Overview

01

Source of Dataset

- BBC News dataset (commonly found on Kaggle or academic repositories).
- Curated collection of 2,225 news articles from BBC.

02

Category Distribution

Five distinct categories:

- Business – 510 articles
- Entertainment – 386 articles
- Politics – 417 articles
- Sport – 511 articles
- Tech – 401 articles

03

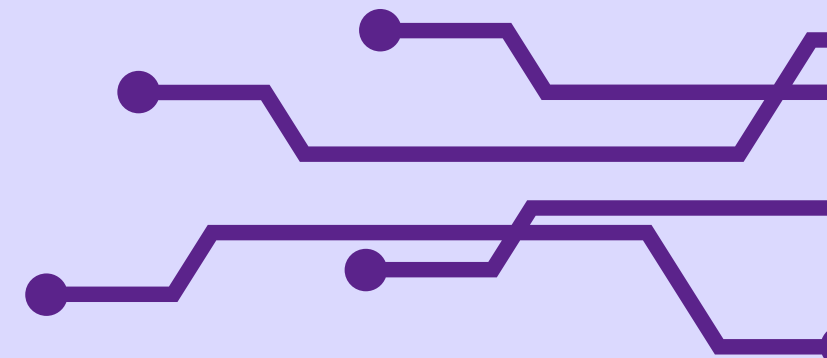
Dataset Format

- Text files organized in subfolders by category.
- Each file: a single news article (raw text).

04

Visualization & Insights

- Pie chart or bar graph showing article count per category.
- Word cloud (optional) to show common words per category.
- Observation: Some overlap in vocabulary between business and tech.



Data Preprocessing



01

Steps

- Lowercasing and punctuation removal
- Tokenization: Splitting text into individual words
- Stopword removal: Eliminating common but uninformative words (e.g., "the", "and")
- Sequence padding: Ensuring uniform input length

02

Tools Used

NLTK, Gensim, Scikit Learn

03

Deployment

Streamlit — Frontend web application for real-time news category prediction



Word2Vec Model Architecture



01

Word2vec Model Architecture

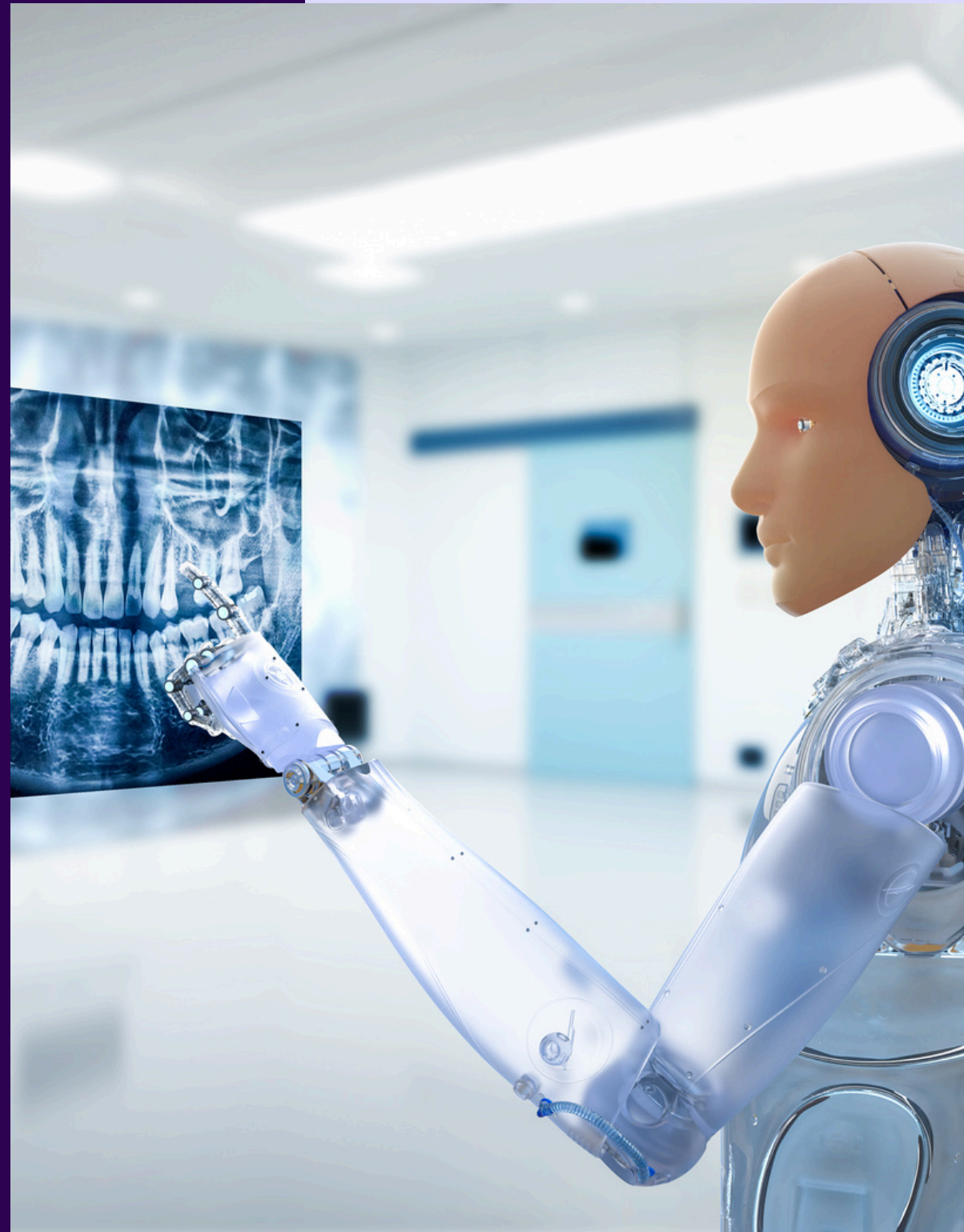
1. Text data is cleaned, tokenized, and lemmatized during preprocessing.
2. Each word is mapped to a dense vector using a trained Word2Vec model.
3. Sentence vectors are generated by averaging word vectors in each article.
4. These vectors are fed into a classifier (e.g., Logistic Regression) to predict the news category.

02

Diagram

Text → Preprocessing → Word2Vec → Vector
Averaging → Classifier → Output Category





Model Compilation & Training



01

The sentence vectors (from Word2Vec) are split into training and test sets.

02

Classification models like Logistic Regression and SVM are trained on these vectors.

03

Model training is performed with 80–20 train–test split using Scikit-learn.

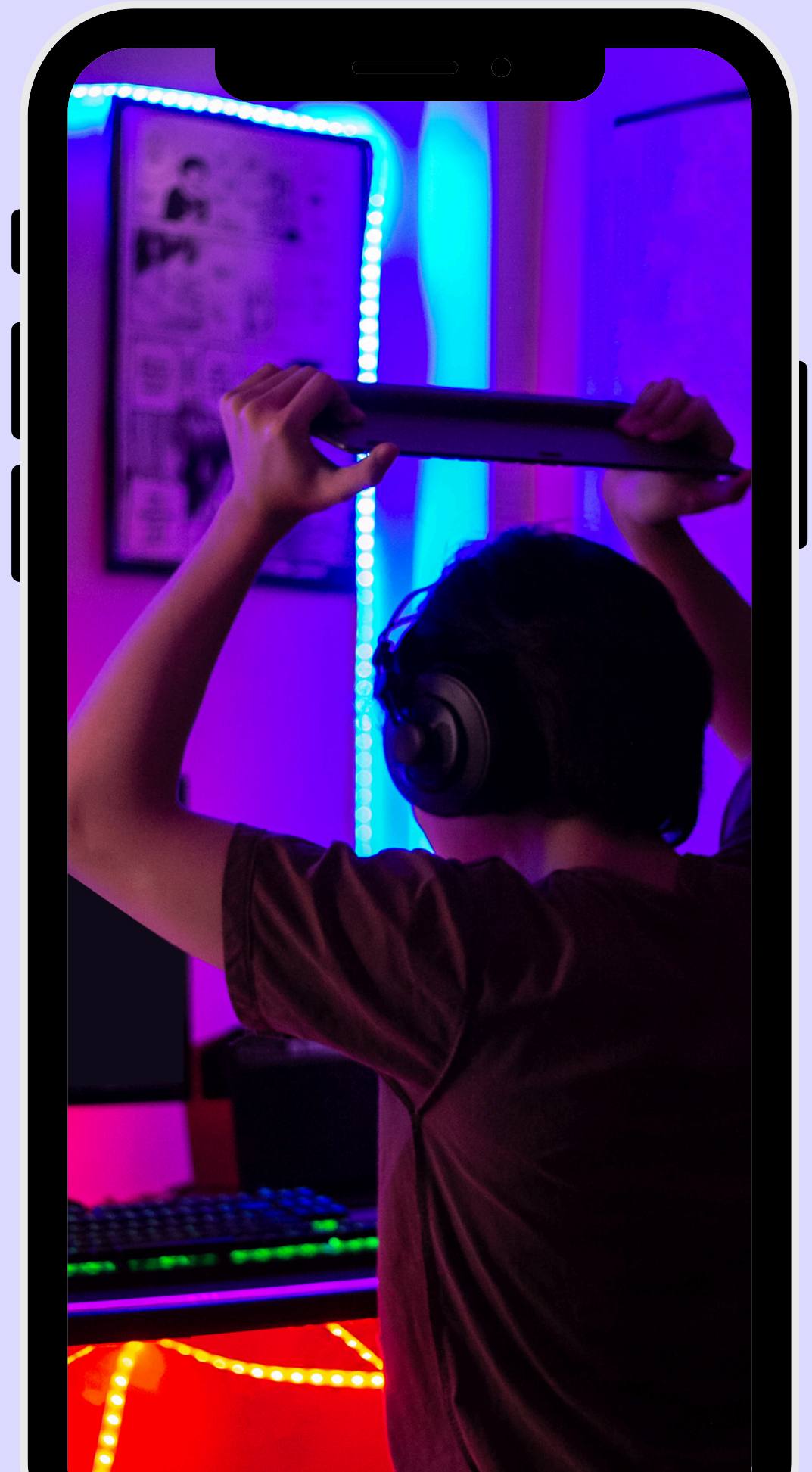
04

Evaluation metrics like accuracy, precision, and recall are computed on the test set.

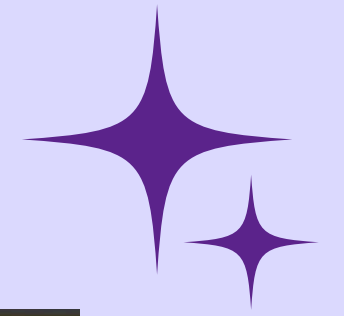
Model Evaluation

Initial Consultation → **Solution Design** → **Implementation & Support**

- Model performance is assessed using Accuracy, Precision, Recall, and F1-Score.
- A Confusion Matrix highlights how well the model distinguishes between categories.
- The classifier achieved high accuracy, especially in Sport and Politics categories.
- Some misclassification occurred between Business and Tech due to vocabulary overlap.



OUTPUT





BBC News Classifier

This app classifies news articles into the following categories:

Business, Entertainment, Politics, Sport, Tech

Please enter the full text of a news article in the box. The classifier will then predict its category.

Deploy ⋮

News Article Classification ↗

Enter News Article Text Here:

Mukesh Kumar sets up comfortable win as DC close the gap up top

DC restricted LSG to just 72 runs in last ten overs, before Rahul and Porel's fifties took visitors home

22-Apr-2025 • 11 hrs ago • Deivarayan Muthu

Markram, Marsh and Badoni push LSG to 159 against DC in Lucknow

Markram, Marsh and Badoni push LSG to 159 against DC in Lucknow

Mukesh Kumar bagged 4 for 33 for DC, who hit back after an 87-run opening stand

22-Apr-2025 • 13 hrs ago • ESPNcricinfo staff

Cricinfo Scotland reaches lamiable resolution with former player Harman Tahir

🚀 Classify

Predicted Category:

sport

Developed with Streamlit and Python



THANK YOU

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